

## *Trends of outpatient prescription drug utilization in US children, 2002-2010.*

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OBJECTIVE: To describe trends in outpatient prescription drug utilization in US children and the changes in major areas of pediatric therapeutic use for the years 2002 through 2010. METHODS: Large prescription databases (the IMS Vector One: National and Total Patient Tracker) were used to examine national drug utilization patterns for the US pediatric population (ages 0-17 years) from 2002 through 2010. RESULTS: In 2010, a total of 263.6 million prescriptions were dispensed to the US pediatric population, 7% lower than in 2002, while prescriptions dispensed to the adult population increased 22% during the same time. Analysis of pediatric drug utilization trends for the top 12 therapeutic areas in 2010 compared with 2002 showed decreases in systemic antibiotics (-14%), allergies (-61%), pain (-14%), depression (-5%), and cough/cold without expectorant (-42%) prescriptions, whereas asthma (14%), attention-deficit/hyperactivity disorder (46%), and contraceptive (93%) prescriptions increased. In 2010, amoxicillin was the most frequently dispensed prescription in infants (aged 0-23 months) and children (aged 2-11 years). Methylphenidate was the top prescription dispensed to adolescents (aged 12-17 years). Off-label use was identified, particularly for lansoprazole; ~358,000 prescriptions were dispensed in 2010 for infants <1 year old. CONCLUSIONS: Changes in the patterns of pediatric drug utilization were observed from 2002 to 2010. Changes include a decrease in antibiotic use and an increase in attention-deficit/hyperactivity disorder medication use during the examined time. This article provides an overview of pediatric outpatient drug utilization, which could set the stage for further in-depth analyses.